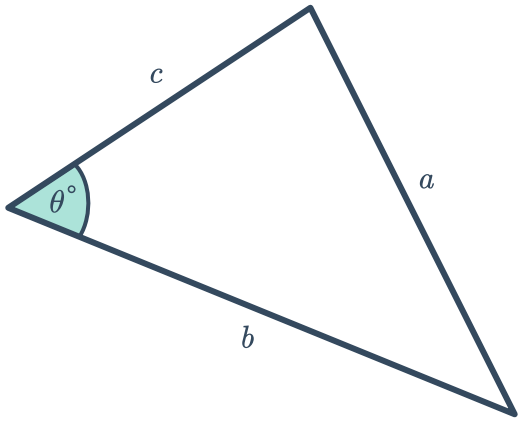
Cosine rule

1 Write an expression for $\cos\theta$ using the cosine rule for the following triangle:

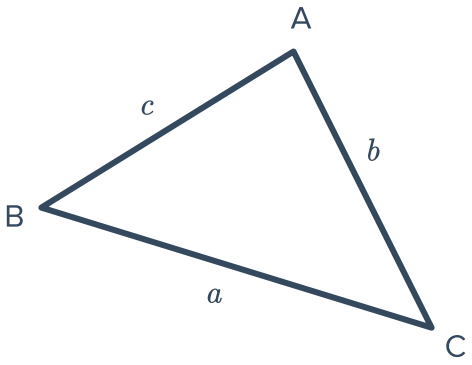


2 Consider the triangle given below:

a Write an expression for $\cos A$ using the cosine rule.

b Write an expression for a^2 using the cosine rule.

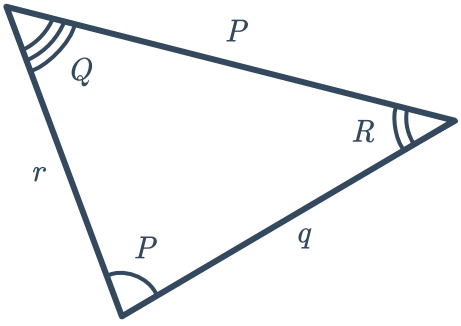
c Write an expression for b using the cosine rule.



3 Consider the triangle given below:

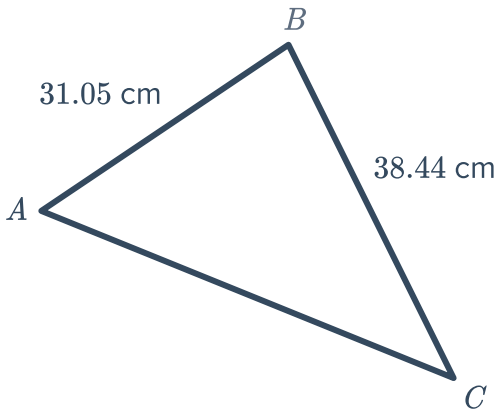
a Write an expression for $\cos Q$ using the cosine rule.

b Write an expression for r^2 using the cosine rule.



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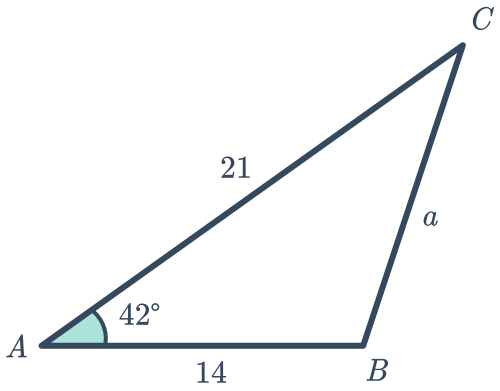
4 To use the cosine rule to find the length of AC, which angle would need to be given?



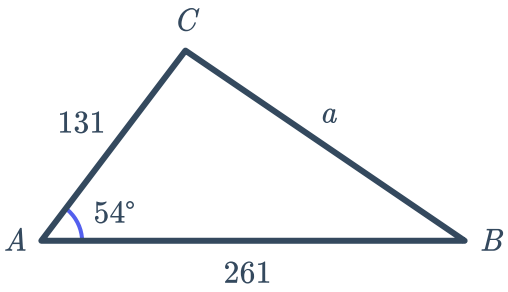
Unknown sides

5 Find the length of the missing side in each of the following triangles using the cosine rule. Round your answers to two decimal places.

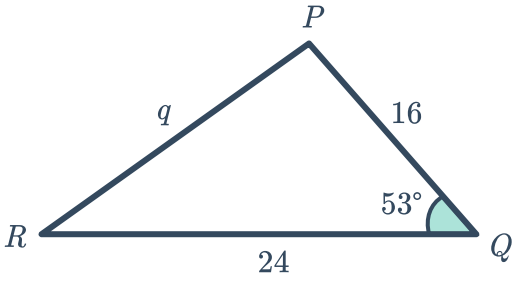
a



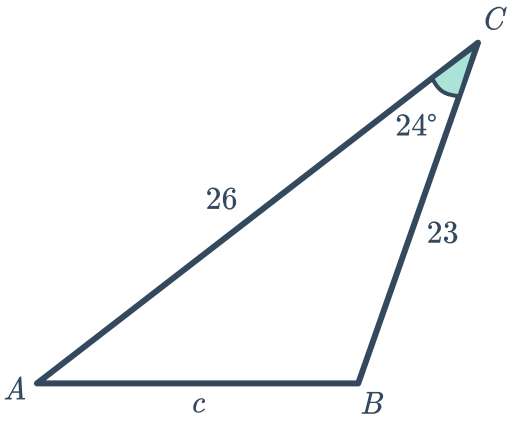
b



c

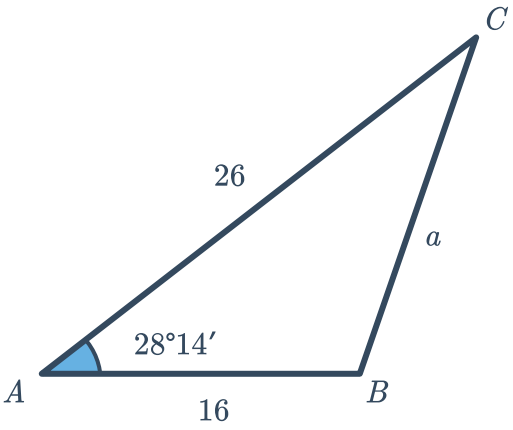


d

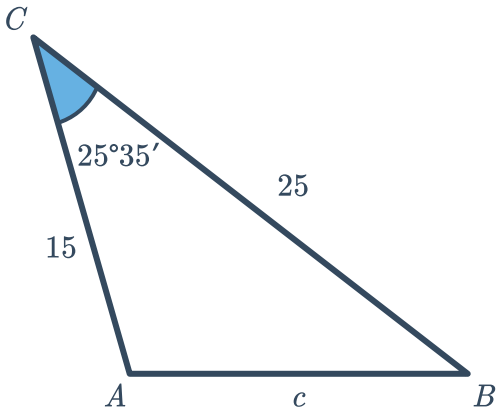


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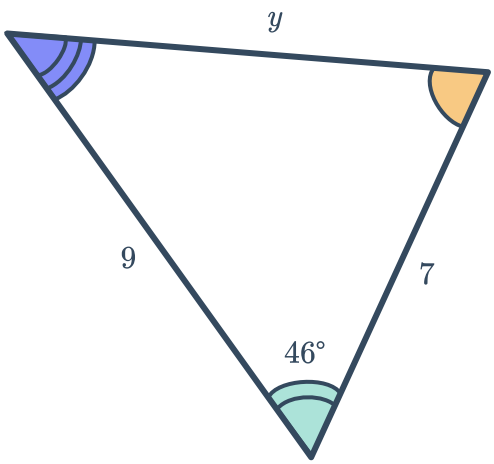
e



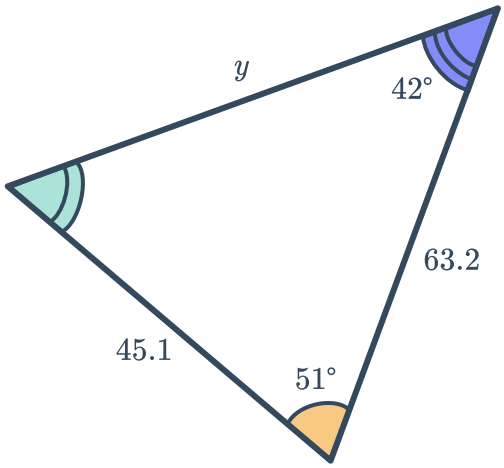
f



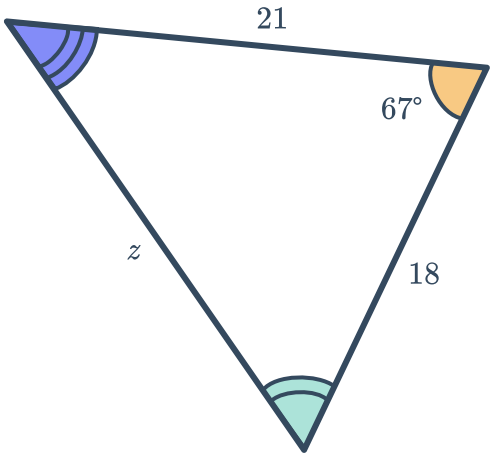
g



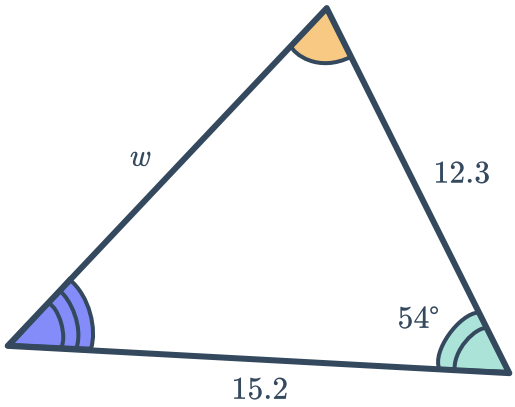
h



i



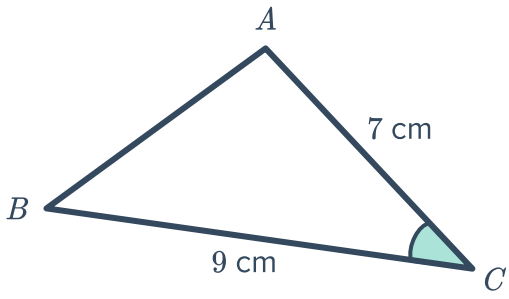
j



6 In $\triangle ABC$, $\cos C = \frac{8}{9}$:

Find the exact length of side AB in centimetres.

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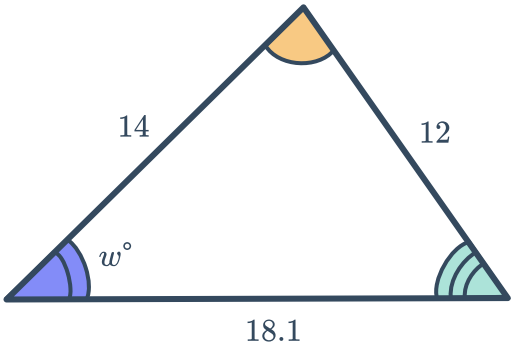


7 In $\triangle QUV$, $q = 5$, $u = 6$ and $\cos V = \frac{3}{5}$. Find the value of v.

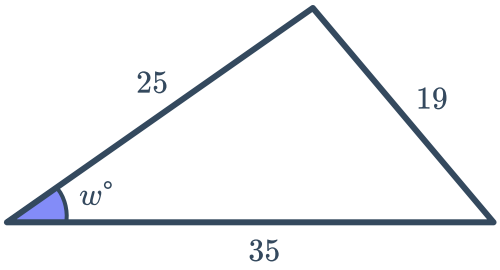
Unknown angles

8 For each of the following triangles, find the value of the pronumeral to the nearest degree:

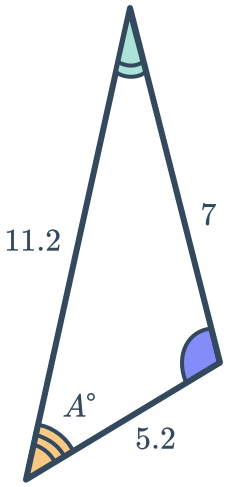
a



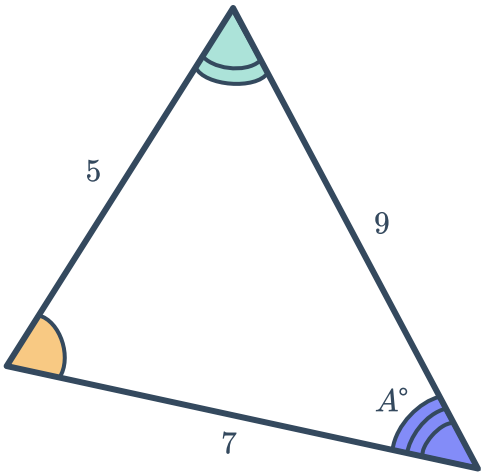
b



c

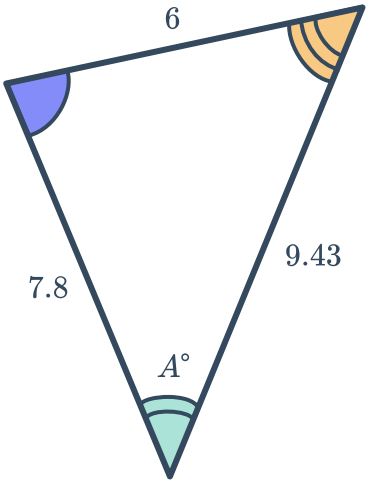


d

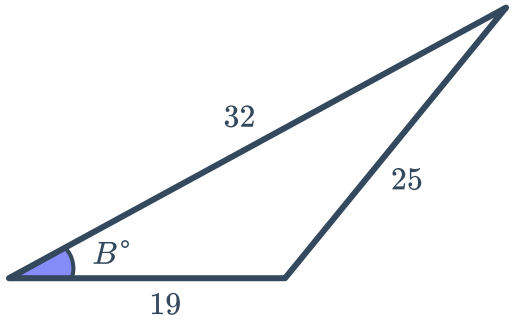


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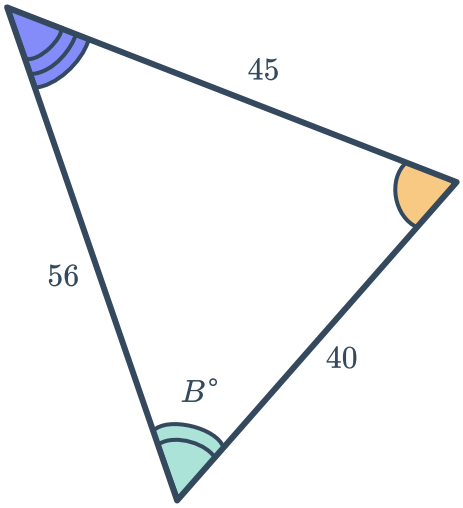
e



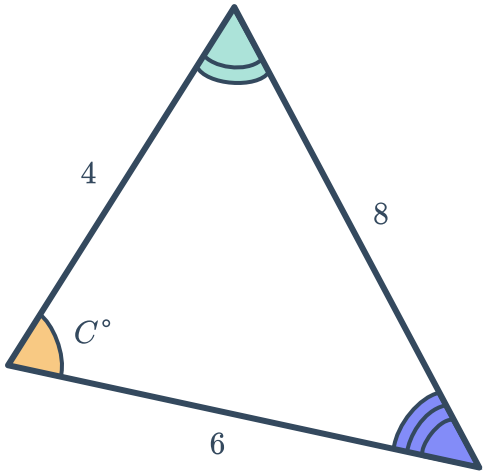
f



g

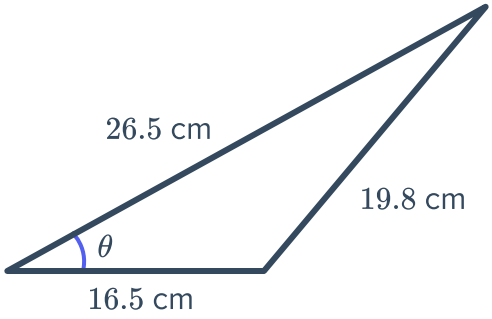


h

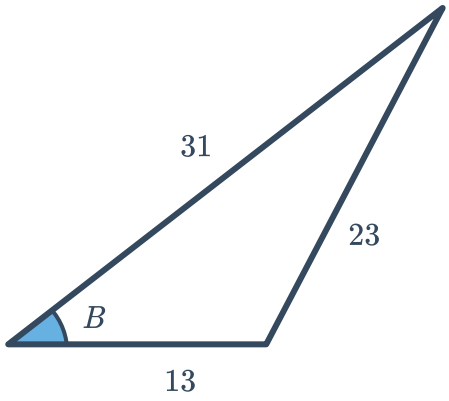


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9 Find the value of θ in the following triangle. Round your answer to the nearest hundredth of a degree.



10 Find the value of B in the following triangle. Round your answer to the nearest second.



11 In $\triangle QUV$, $v = 8$, $u = 9$ and $q = 15$. Solve for $\cos Q$.

12 The sides of a triangle are in the ratio $4 : 5 : 8$. Find θ, the smallest angle in the triangle, to the nearest degree.

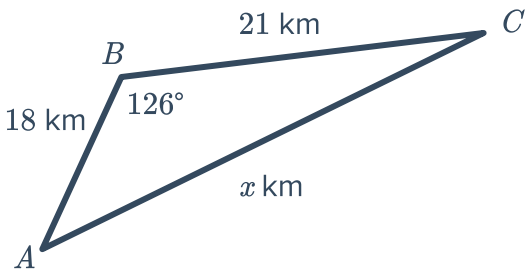
13 A triangle has sides of length 11 cm, 18 cm and 8 cm. Find x, the largest angle in the triangle, to the nearest degree.

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Applications

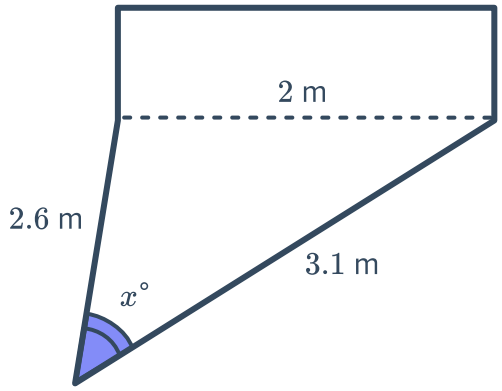
14 Mae went for a bike ride on Sunday morning from Point A to Point B, which was $18\ \text{km}$ long. She then took a 126° turn and rode from Point B to Point C, which was $21\ \text{km}$ long.

Find x, the distance in kilometres from her starting point to Point C to two decimal places.



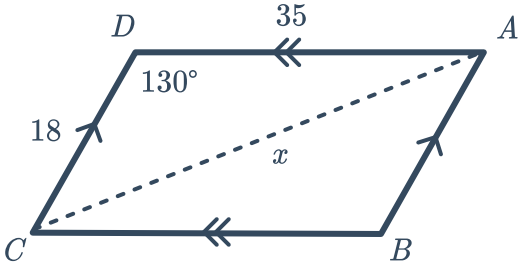
15 A goal has posts that are $2\ \text{m}$ apart. Buzz shoots for the goal when he is $2.6\ \text{m}$ from one post and $3.1\ \text{m}$ from the other post.

Find the size of the angle, x, in which he can score a goal. Round your answer to the nearest degree.



16 Find the length of the diagonal, x, in parallelogram $ABCD$.

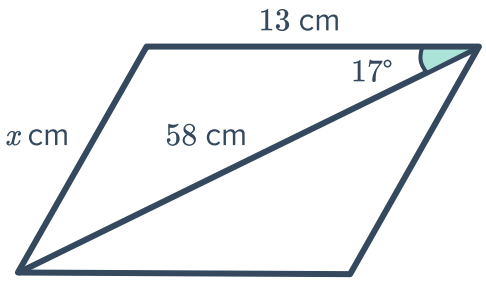
Round your answer to two decimal places.



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17 Consider the parallelogram in the given diagram that has a side of length $13\ \text{cm}$ and a diagonal of length $58\ \text{cm}$:

Find the value of x. Round your answer to one decimal place.



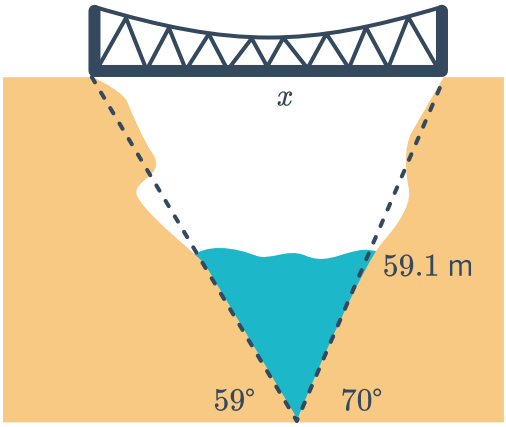
18 In a sailing boat race, teams must start at buoy A and navigate around buoys B and C before returning to buoy A to cross the line. The first leg of the race is $100.6\ \text{km}$ long, the second leg of the race is $190.3\ \text{km}$ long, and the angle between these legs is 143°.

a Find x, the distance of the third leg of the race, correct to one decimal place.

b Hence, find the total length of the race, correct to one decimal place.

19 A bridge connects two towns on either side of a gorge, where one side of the gorge is inclined at 59° and the other side is inclined at 70°. The length of the steeper incline is $59.1\ \text{m}$.

Find x, the length of the bridge. Round your answer correct to one decimal place.



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